DIGITAL IMAGE AND VIDEO PROCESSING COURSE PROJECT REPORT (IMP302)

TOPIC: MULTI-DEGRADATION LICENSE PLATE IMAGE RESTORATION COMBINED WITH CHARACTER SEGMENTATION AND RECOGNITION USING CLASSICAL IMAGE PROCESSING METHODS

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Abstract

Two-line motorcycle license plates captured by outdoor traffic and parking surveillance cameras are frequently degraded by linear motion blur, low-light sensor noise, bipolar impulse corruption, periodic electromagnetic interference, and severe perspective foreshortening. Because adjacent glyphs on a canonical 160 × 128 rectified plate are separated by only 3--4 pixels, even moderate directional blur (L = 7 pixels) bridges the inter-character gap, causing connected-component segmentation to collapse and reducing full-plate recognition accuracy to 0.00%. We study a fully transparent, white-box classical image processing and statistical pattern recognition pipeline implemented from first principles without deep neural networks. The proposed architecture comprises four stages: (i) a multi-degradation restoration stage combining 2D Haar wavelet BayesShrink denoising with a hybrid Wiener–Constrained Least Squares (CLS) deconvolution filter, a 3 × 3 center-weighted median filter for impulse noise, and a conjugate Butterworth notch reject filter for periodic interference; (ii) a three-stream morphological and vertical-Sobel detector with minimum-area rotated rectangle scoring, IoU-based Non-Maximum Suppression (NMS) ordered strictly left-to-right, and Douglas–Peucker 4-point perspective homography rectification; (iii) a shadow-resilient local Gaussian adaptive binarizer paired with a maximum contiguous run-length border suppression algorithm that preserves aligned top character caps; and (iv) a 480-dimensional hybrid descriptor fusing sequency-ordered 2D Walsh–Hadamard Transform (WHT) coefficients, Histograms of Oriented Gradients (HOG), 7 Hu moment invariants, boundary Fourier descriptors, region covariance eccentricity, and Euler topological numbers (E = C - H), evaluated across seven classical classifiers with topological post-verification. On a zero-leakage plate-level benchmark (80 training, 17 validation, and 18 test plates expanded 6× across six physical degradation regimes to 690 vehicle images and 5,643 character patches), Stage 1 restoration yields up to +17.98 dB PSNR gain, recovers motion-blurred character recognition on the held-out test split from 3.85% to 93.59%, and reaches 96.99% 5-fold cross-validation character accuracy. We also audit ground-truth label discrepancies in the raw corpus, evaluate four challenging in-the-wild stress-test scenes, and analyze the remaining failure modes.

Keywords: Image restoration, Wiener deconvolution, constrained least squares, Haar wavelet shrinkage, perspective homography, Otsu thresholding, K-Means segmentation, Walsh–Hadamard transform, boundary Fourier descriptors, Euler topology, license plate recognition.

1 Introduction

Automatic License Plate Recognition (ALPR) is a core component of urban traffic monitoring, automated parking management, and law enforcement surveillance [1, 2]. While single-line passenger car plates in controlled highway lanes are well studied, Vietnamese two-line motorcycle plates (190 mm × 140 mm physical dimensions containing 8 or 9 alphanumeric characters arranged across an upper 4-character row and a lower 4- or 5-digit row) present distinct signal processing and geometric challenges. In low-cost parking and street cameras, vehicles are often photographed at oblique angles ($\pm 30^{\circ}$ in-plane roll and out-of-plane yaw), under non-uniform shadow gradients, or while moving through the camera exposure window.

Let $f(x,y) \in R^{M \times N}$ denote an ideal, uncorrupted surveillance frame. Under a linear space-invariant (LSI) optical degradation model coupled with projective viewing geometry T_H and additive or impulsive sensor noise $\eta(x,y)$, the observed frame $g(x,y)$ is given by [3]:

$$g(x,y) = \mathcal{T}_{\mathbf{H}}\{(h * f)(x,y)\} + \eta(x,y), \tag{1}$$

where $h(x,y)$ is the 2D Point Spread Function (PSF) induced by relative motion between the vehicle and the imaging sensor, $*$ denotes 2D spatial convolution, and $H \in R^{3 \times 3}$ is a projective homography matrix. Recovering the exact character sequence $\hat{s} = (\hat{c}_1, \hat{c}_2, ..., \hat{c}_L)$ from $g(x,y)$ requires inverting both the radiometric degradation (h, η) and the geometric distortion T_H prior to connected-component isolation and statistical pattern classification.

Surveys of classical license plate analysis [1, 2] and our empirical audit of degraded surveillance frames reveal three primary causes of pipeline failure:

1. **Coupled blur–noise amplification and stroke coalescence:** On a rectified 160×128 plate image, the horizontal spacing between neighboring glyphs is only 3--4 pixels. A linear motion blur kernel of length $L = 7$ pixels smears foreground ink across this gap, merging four or five digits into a single connected component. Naive frequency-domain inversion ($\hat{F}(u,v) = G(u,v)/H(u,v)$) amplifies high-frequency noise $\eta(x,y)$ near the spectral zeros of the sinc transfer function $H(u,v)$, destroying stroke legibility.
2. **Oblique foreshortening and border-induced cap truncation:** When a motorcycle is viewed from the side or at a steep tilt, the axis-aligned bounding box of the plate exhibits low rectangular extent (Extent < 0.65) and distorted aspect ratio ($W < H$). Furthermore, conventional border removal heuristics that zero out horizontal scanlines exceeding 40% foreground density inadvertently slice off the aligned top horizontal caps of characters such as `'2'`, `'9'`, `'D'`, and `'7'`, turning a closed `'D'` into an open `'U'`-shaped artifact.
3. **Structural ambiguity and severe class imbalance:** In the Vietnamese license plate typeface, pairs such as `'D'` vs. `'B'`, `'E'` vs. `'C'`, and `'7'` vs. `'1'` share nearly identical outer bounding boxes and vertical projections. At the same time, rare series letters (`'A'`, `'D'`, `'E'`, `'R'`, `'T'`, `'Y'`) have very few or zero instances in standard 100-plate benchmarks, causing probability calibration (Platt scaling) in margin classifiers to degenerate.

Research gap. Modern ALPR deployments increasingly rely on deep convolutional or Transformer detectors (e.g., YOLO-based plate detectors paired with optical character recognition networks) [4, 5]. While effective on large GPU-trained corpora, black-box neural pipelines obscure the mathematical relationship between frequency-domain deconvolution, morphological geometry, and topological feature invariance, and degrade unpredictably when tested on out-of-distribution sensor corruptions (such as periodic power-line interference or high-density impulse noise). Conversely, existing classical ALPR studies typically evaluate only a single stage (such as Otsu binarization or template matching) on clean, fronto-parallel crops without rigorous plate-level train/validation/test partitioning or multi-degradation restoration. How classical spatial-frequency restoration interacts with projective homography rectification, contiguous run-length segmentation, and hybrid transform–topological classification in a complete end-to-end system remains insufficiently documented.

Contributions. This paper designs, implements, and audits a complete four-stage classical license plate restoration and recognition pipeline. Specifically:

1. We formulate a multi-degradation restoration stage (Section 3.2) that pairs 2D Haar wavelet BayesShrink denoising with a hybrid Wiener–Constrained Least Squares (CLS) deconvolution filter, alongside a 3×3 center-weighted median filter and a conjugate Butterworth notch reject filter. Restoration improves PSNR by $+2.35$ to $+17.98$ dB on four of five physical degradation regimes (AWGN, impulse, periodic, motion blur) and rescues held-out test character recognition under motion blur from 3.85% to 93.59%.
2. We develop a three-stream morphological detector with minimum-area rotated rectangle scoring, IoU-based Non-Maximum Suppression (NMS) ordered strictly left-to-right, and Douglas–Peucker 4-point perspective homography rectification (Section 3.3), coupled with a maximum contiguous run-length (`_max_run`) border filter (Section 3.4) that eliminates frame borders and `"VIE"` flag sticker dividers without truncating character caps.
3. We construct a 480-dimensional hybrid feature representation combining sequency-ordered 2D Walsh–Hadamard Transform (WHT) coefficients, HOG descriptors, boundary Fourier descriptors, region covariance eccentricity, 7 Hu moment invariants, and Euler topological numbers ($E = C - H$), and benchmark seven classical classifiers with topological post-verification for structurally ambiguous glyph pairs (`'D'` vs. `'B'`, `'E'` vs. `'C'`, `'7'` vs. `'1'`) (Section 3.5).
4. We establish a zero-leakage stratified plate-level evaluation protocol (80 train / 17 validation / 18 test plates expanded 6× to 690 images and 5,643 character patches), audit and rectify ground-truth labeling errors in the raw dataset, and validate generalization on four challenging in-the-wild stress-test scenes and real-time surveillance video streams (Sections 4 and 5).

We do not claim superiority over large-scale GPU-trained deep learning systems on unconstrained open-world benchmarks. Our goal is to characterize a fully specified, mathematically interpretable classical pipeline from first principles and to isolate the

quantitative contribution of each restoration, segmentation, and classification component.

2 Related Work

2.1 License plate recognition systems

Table 1 positions our pipeline relative to representative classical and hybrid ALPR methodologies in the literature. Early ALPR systems relied on vertical edge projection and global thresholding followed by pixel-level template matching [1, 2]. While computationally light, projection-based methods require near-zero roll angle and uniform illumination. Subsequent classical pipelines introduced morphological top-hat/black-hat operators [6] and Connected Component Analysis (CCA) [7], paired with Support Vector Machines (SVM) on hand-crafted gradient features [8]. More recently, one-stage deep detectors [4, 5] have achieved high accuracy on large datasets at the cost of millions of learned parameters and GPU inference requirements. In contrast, our work retains a 100% white-box classical formulation while addressing the four failure modes that limit traditional pipelines: coupled motion-blur/noise restoration, 4-point projective homography for foreshortened plates, contiguous run-length border suppression, and topological glyph disambiguation.

Table 1: Design elements of representative license plate recognition pipelines compared with the present work. "—" indicates that the component is not part of the method's stated formulation.

Method	Paradigm	Degraded Image Restoration	Geometric Rectification	Binarization & Border Filtering	Feature & Classifier Design
Anagnostopoulos et al. [1]	Classical	—	Connected-component tilt	Sliding concentric windows	Pixel projections + PNN
Du et al. [2]	Classical survey	—	Hough line deskewing	Global Otsu thresholding	Template matching / ANN
Azam & Islam [6]	Classical	Spatial median only	Affine rotation	Local adaptive thresholding	Freeman chain code + k-NN
Dalal & Triggs + SVM [8, 16]	Classical ML	—	—	Global Otsu + density cut	HOG + Linear/RBF SVM
Silva & Jung (WPOD-Net) [5]	Deep CNN	Implicit (data aug.)	Learned planar affine warp	End-to-end CNN detector	YOLO-based OCR net
This work	**Classical + ML**	**2D Haar DWT + Hybrid Wiener–CLS, Weighted Median, Butterworth Notch**	**3-stream Rotated Extent + 4-pt Perspective Homography**	**Local Gaussian Adaptive + Max Contiguous Run-Length (`_max_run`)**	**480-D Hybrid (WHT + HOG + Fourier + Cov + Hu + Euler) + 7 Classifiers + Topo Rules**

2.2 Spatial, frequency, and wavelet image restoration

Classical image restoration models degradation in either the spatial domain (for additive and impulsive noise) or the frequency domain (for periodic interference and convolutional blur) [3]. Order-statistic filters—most notably the median filter and the center-weighted median filter introduced by Ko and Lee [9]—suppress bipolar salt-and-pepper impulses while preserving step edges and narrow stroke corners better than linear arithmetic or geometric mean smoothers. For skewed positive noise distributions such as Rayleigh corruption, harmonic and contraharmonic mean filters provide targeted attenuation [3]. Periodic sinusoidal interference concentrates energy into conjugate impulse pairs in the 2D Discrete Fourier Transform (DFT) spectrum, allowing selective removal via Butterworth notch reject filters [3].

When an image is blurred by a known or estimated motion PSF $h(x,y)$ in the presence of additive noise, direct inverse filtering fails near the zeros of the optical transfer function $H(u,v)$. Wiener deconvolution [10] minimizes the statistical mean-square error using the noise-to-signal power ratio K, whereas Hunt's Constrained Least Squares (CLS) filter [11] minimizes the second-order Laplacian roughness $\|p * \hat{f}\|^2$ subject to a noise norm constraint. To prevent residual high-frequency Gaussian noise from being amplified during frequency-domain inversion, Donoho and Johnstone [12] and Chang et al. [13] established wavelet shrinkage (BayesShrink) in the 2D Discrete Wavelet Transform (DWT) domain. Our Stage 1 pipeline unifies these principles by cascading 2D Haar BayesShrink denoising with a hybrid Wiener–CLS deconvolution transfer function.

2.3 Geometric rectification and character segmentation

Detecting license plates in cluttered vehicle scenes commonly leverages the high horizontal frequency of vertical character strokes on a bright reflective background [1, 6]. Morphological black-hat filtering highlights dark foreground strokes against bright backgrounds, while vertical Sobel gradients $|G_x|$ capture stroke transitions. Standard axis-aligned bounding boxes degrade when the camera roll or yaw angle exceeds $\pm\ 15^{\circ}$. By evaluating the minimum-area rotated bounding rectangle (`minAreaRect`) and approximating the plate contour via the Douglas–Peucker polygonal reduction algorithm [14], a 4-point planar homography $H \in R^{3\times 3}$ can warp foreshortened quadrilaterals into a fronto-parallel canonical grid using bicubic convolution interpolation [15].

For character binarization, Otsu's method [7] maximizes between-class gray-level variance globally, whereas MacQueen's K-Means clustering (K=2) [16] iteratively partitions pixel intensities by minimizing within-cluster sum-of-squares. Under diagonal vehicle shadows, global thresholds merge shadowed digits with the background; local Gaussian adaptive thresholding [17] overcomes this by estimating a spatially varying background surface. Once binarized, connected components are extracted via two-pass 8-connected labeling [18].

2.4 Transform-domain and topological descriptors

Representing normalized 32×32 character patches with compact, discriminative features is central to statistical pattern recognition [19]. Orthogonal 2D transforms—including the Discrete Hartley Transform (DHT) [20], the Discrete Cosine Transform (DCT-II) [21], and the sequency-ordered Walsh–Hadamard Transform (WHT) [22]—concentrate low-spatial-frequency stroke energy into the top-left transform coefficients. Boundary and regional shape descriptors complement transform spectra: Zahn and Roskies [23] parameterized closed contours via 1D DFT Fourier descriptors; Hu [24] derived seven algebraic moment invariants orthogonal to translation, scale, and in-plane rotation; and topological descriptors such as the Euler characteristic $E = C - H$ (connected components minus holes) [3, 18] capture structural invariants that remain unchanged under non-linear elastic deformations. Combining these global/topological cues with local Histograms of Oriented Gradients (HOG) [8] provides a rich representation for classical minimum-distance, correlation, Gaussian Bayes, multi-layer perceptron (MLP), k-Nearest Neighbors (k-NN), Random Forest, and RBF-kernel SVM classifiers [19, 25].

3 Proposed Method

3.1 Overview

Given an input BGR vehicle frame $I \in R^{H\times W\times 3}$ (or a sequence of frames from a video stream), our system outputs a set of detected license plates $\{(b_m, q_m, \hat{s}_m, c_m)\}_{m=1}^{M}$ ordered strictly from left to right along the horizontal image axis, where $b_m = (x_m, y_m, w_m, h_m)$ is the bounding box, $q_m \in R^{4\times 2}$ is the 4-vertex quadrilateral, $\hat{s}_m$ is the recognized license plate string formatted to Vietnamese syntax (e.g., `29-D2 399.99`), and $c_m \in [0, 1]$ is the mean character posterior confidence. Table 2 summarizes the four sequential stages, their input/output resolutions, and their core mathematical operators.

Table 2: Summary of the four stages in the proposed classical license plate restoration and recognition pipeline.

Stage	Module (`src/`)	Input Representation	Output Representation	Core Mathematical Operators
Stage 1	`restoration.py`	Degraded frame $g(x,y) \in R^{H\times W\times 3}$	Restored frame $\hat{f}(x,y) \in R^{H\times W\times 3}$	3× 3 Center-Weighted Median, Harmonic Mean, Butterworth Notch $H_{NR}(u,v)$, 2D Haar DWT BayesShrink + Hybrid Wiener–CLS Deconvolution
Stage 2	`detection.py`	Restored frame $\hat{f}(x,y)$	Rectified plate ROIs $\{\hat{f}_{roi}^{(m)}\} \in R^{160\times 128\times 3}$	Black-Hat $(f \bullet B)-f$, Vertical Sobel $\|G_x\|$, 3-Stream Rotated Extent Scoring, NMS (IoU ≤ 0.25), Douglas–Peucker 4-Point Homography $H_{3\times 3}$
Stage 3	`segmentation.py`	Rectified plate $\hat{f}_{roi} \in R^{160\times 128\times 3}$	Ordered character crops $\{I_k\}_{k=1}^{L} \in R^{32\times 32}$	Otsu / K-Means (K=2) / Local Gaussian Adaptive Thresholding, Max Contiguous Run-Length Border Filter (`_max_run`), 8-Connected Labeling
Stage 4	`recognition.py`	Character patches $\{I_k\}_{k=1}^{L} \in R^{32\times 32}$	Recognized string $\hat{s}$ & confidence c	2D WHT/DCT/DHT (100-D) + HOG (324-D) + Hu/Cov/Fourier/Euler (56-D), PCA (80-D), 7 Classical Classifiers + Topological Post-Verification

3.2 Stage 1: Multi-degradation image restoration

To handle diverse real-world corruptions without degrading clean frames, Stage 1 implements dedicated spatial, frequency, and wavelet-deconvolution operators on the luminance channel Y in the YCrCb color space.

Spatial order-statistic and mean filters. For additive Gaussian, skewed Rayleigh, and bipolar impulse corruptions over an $m \times n$ neighborhood S_{xy}, we implement the arithmetic, geometric, harmonic, and contraharmonic mean filters [3]:

$$\hat{f}_{\text{harm}}(x,y) = \frac{mn}{\sum_{(s,t)\in S_{xy}} \frac{1}{g(s,t)+\epsilon}}, \qquad \hat{f}_{\text{contra}}(x,y) = \frac{\sum_{(s,t)\in S_{xy}} g(s,t)^{Q+1}}{\sum_{(s,t)\in S_{xy}} g(s,t)^{Q}}. \tag{2}$$

Here $\hat{f}_{\text{harm}}(x,y)$ effectively attenuates positive-skewed Rayleigh noise while preserving dark character ink. For bipolar salt-and-pepper noise ($P_a = P_b = 0.02$), a standard unweighted 3×3 median filter rounds off the acute vertices of thin digits (`'1'`, `'7'`, `'4'`). We therefore employ a 3×3 center-weighted median filter [9] with integer replication mask W_{WMF}:

$$\mathbf{W}_{\text{WMF}} = \begin{vmatrix} 1 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 1 \end{vmatrix}, \qquad \hat{f}_{\text{WMF}}(x,y) = \text{median}\{g(s,t),\ \ldots, g(s,t)_{W_{\text{WMF}}(s,t)\text{times}} | (s,t) \in S_{xy}\}. \tag{3}$$

Frequency-domain Butterworth notch reject filtering. Periodic sinusoidal interference $\eta(x,y) = A\sin(2\pi(u_0\, x/M + v_0\, y/N))$ produces conjugate impulse spikes at $(M/2 \pm u_k, N/2 \pm v_k)$ in the centered 2D DFT magnitude spectrum $|G(u,v)|$. We suppress Q spike pairs using a symmetric Butterworth notch reject transfer function of order $n=2$ and cutoff radius $D_{0k}=6.0$:

$$H_{\text{NR}}(u,v) = \prod_{k=1}^{Q} \left[\frac{1}{1+\left(\frac{D_{0k}}{D_k(u,v)}\right)^{2n}}\right]\left[\frac{1}{1+\left(\frac{D_{0k}}{D_{-k}(u,v)}\right)^{2n}}\right], \tag{4}$$

where $D_{\pm k}(u,v) = \sqrt{(u - M/2 \mp u_k)^2 + (v - N/2 \mp v_k)^2}$, and reconstruct the clean spatial image via $\hat{f}(x,y) = \text{Re}\{F^{-1}\{H_{\text{NR}}(u,v)G(u,v)\}\}$.

2D Haar wavelet BayesShrink and hybrid Wiener–CLS deconvolution. For the coupled degradation $g(x,y) = (h * f)(x,y) + \eta(x,y)$ with linear motion PSF $h(x,y)$ of length L and angle θ, applying a direct inverse filter $\hat{F}(u,v) = G(u,v)/H(u,v)$ amplifies the noise term $N(u,v)/H(u,v)$ whenever $|H(u,v)| \to 0$. We decouple noise suppression from deconvolution in two steps:

1. *Wavelet shrinkage:* The luminance channel is decomposed via a 1-level separable 2D Haar DWT (`dwt2_haar`) with low-pass filter $h_\varphi = \frac{1}{\sqrt{2}}[1, 1]$ and high-pass filter $h_\psi = \frac{1}{\sqrt{2}}[1, -1]$ into subbands $\{LL_1, LH_1, HL_1, HH_1\}$. The noise standard deviation $\hat{\sigma}_\eta$ is estimated from the diagonal detail subband HH_1 via Donoho's robust median estimator [12], and each detail subband $s \in \{LH_1, HL_1, HH_1\}$ is soft-thresholded using BayesShrink [13]:

$$\hat{\sigma}_\eta = \frac{\text{median}(|HH_1|)}{0.6745}, \qquad T_s = \frac{\hat{\sigma}_\eta^2}{\sqrt{\max(\sigma_s^2 - \hat{\sigma}_\eta^2, 0) + \epsilon}}, \tag{5}$$

$$\tilde{d}_s(x,y) = \text{sign}(d_s(x,y))\max(|d_s(x,y)| - T_s, 0). \tag{6}$$

Inverting the Haar transform (`idwt2_haar`) yields a pre-denoised observation $g_{\text{den}}(x,y)$ with 2D FFT $G_{\text{den}}(u,v)$.

2. *Hybrid Wiener–CLS deconvolution:* Let $H(u,v) = F\{h(x,y)\}$ denote the optical transfer function and $P(u,v) = F\{p(x,y)\}$ denote the 2D FFT of the discrete Laplacian operator $p(x,y) = [0, -1, 0; -1, 4, -1; 0, -1, 0]$. Combining Wiener's minimum MSE regularizer K [10] with Hunt's smoothness regularizer $\gamma_{\text{CLS}}|P(u,v)|^2$ [11], we formulate the unit-DC-gain hybrid transfer function:

$$\hat{F}(u,v) = \left[\frac{(1+K)H^*(u,v)}{|H(u,v)|^2 + K + \gamma_{\text{CLS}}|P(u,v)|^2}\right] G_{\text{den}}(u,v), \tag{7}$$

where (1+K) preserves the zero-frequency mean luminance ($\hat{F}(0,0) = G_{den}(0,0)$ since $H(0,0)=1$ and $P(0,0)=0$) and $\gamma_{CLS}|P(u,v)|^2$ damps high-frequency ringing around sharp character boundaries. Finally, unsharp masking (highboost filtering with $k_{boost} = 0.45$) restores stroke edge acutance.

Figure 1: Stage 1 Restoration and Deconvolution Pipeline.

Figure 2: Multi-Noise Degradation Models and Restoration Filters.

3.3 Stage 2: Multi-plate morphological detection and 4-point homography

Morphological and gradient energy map. Given a restored grayscale frame $I \in [0, 255]^{H \times W}$ after full-scale histogram stretching, a rectangular structuring element $B_{25 \times 25}$ computes the black-hat transform $f_{bh} = (I \bullet B_{25 \times 25}) - I$, isolating dark character strokes on bright reflective plates. Simultaneously, a 3×3 horizontal Sobel kernel extracts vertical stroke edges $|G_x| = |\partial I / \partial x|$. The fused energy map $E = 0.5|G_x| + 0.5f_{bh}$ is binarized via Otsu's threshold and processed with morphological closing (17×11) followed by opening (5×5) to form compact candidate blobs.

Three-stream candidate generation with rotated extent. Single-stream contour detectors fail when plates are tilted by $\pm 30^{\circ}$, foreshortened near image borders, or accompanied by bright shop logos. Our candidate generator `_gather_plate_candidates()` pools regions from three complementary streams: (i) multi-threshold external contours ($t \in \{110, ..., 200, t_{Otsu}\}$) for fronto-parallel plates ($1.0 \leq W/H \leq 1.62$, axis-aligned Extent > 0.68); (ii) connected components from the morphologically closed energy map

E_{closed}; and (iii) high-brightness contours ($t \in [135, 210]$) evaluated by their minimum-area rotated bounding rectangle (`cv2.minAreaRect`) with dimensions (W_{rot}, H_{rot}). For a plate tilted at -28.6°, the axis-aligned extent drops to 0.51, whereas the rotated extent remains high:

$$\text{Extent}_{\text{rot}}(C) = \frac{\text{Area}(C)}{W_{\text{rot}} \cdot H_{\text{rot}}} > 0.66, \qquad \text{Aspect}_{\text{rot}}(C) = \frac{\max(W_{\text{rot}}, H_{\text{rot}})}{\min(W_{\text{rot}}, H_{\text{rot}})} \in [1.05, 1.85]. \tag{8}$$

Each candidate region R is scored by combining internal stroke energy and geometric fidelity to the canonical Vietnamese motorcycle plate aspect ratio (1.24):

$$\text{Score}(R) = 1.2\mu_{|G_x|}(R) + 1.6\mu_{\text{bh}}(R) + 1.1\sigma_I(R) + 45.0\text{Extent}_{\text{eff}} - 45.0|\text{Aspect}_{\text{eff}} - 1.24|. \tag{9}$$

Non-Maximum Suppression and strict left-to-right ordering. Overlapping candidates are pruned via Non-Maximum Suppression with threshold $\text{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|} > 0.25$. In multi-vehicle scenes, all surviving plate candidates are verified by downstream character count (≥ 6 valid character components, rejecting circular shop logos) and sorted strictly by their horizontal origin x_m from left to right (`key=lambda p: p["bbox"][0]`).

4-point perspective homography rectification. When a plate is foreshortened ($W < H$) or strongly tilted (Extent ≤ 0.68), affine rotation alone cannot remove projective keystone distortion. We apply Douglas–Peucker polygonal approximation [14] to the convex hull of contour C to extract the four corner vertices $q_i = (x_i, y_i)^T$ ordered clockwise *(top-left, top-right, bottom-right, bottom-left)* and solve for the projective homography $H \in R^{3 \times 3}$ mapping q_i to the canonical target corners $p_i \in \{(0,0), (159,0), (159,127), (0,127)\}$:

$$s_i \begin{vmatrix} u_i \\ v_i \\ 1 \end{vmatrix} = \begin{vmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & 1 \end{vmatrix} \begin{vmatrix} x_i \\ y_i \\ 1 \end{vmatrix}, \tag{10}$$

resampling the rectified 160×128 plate via 4×4 bicubic convolution interpolation [15].

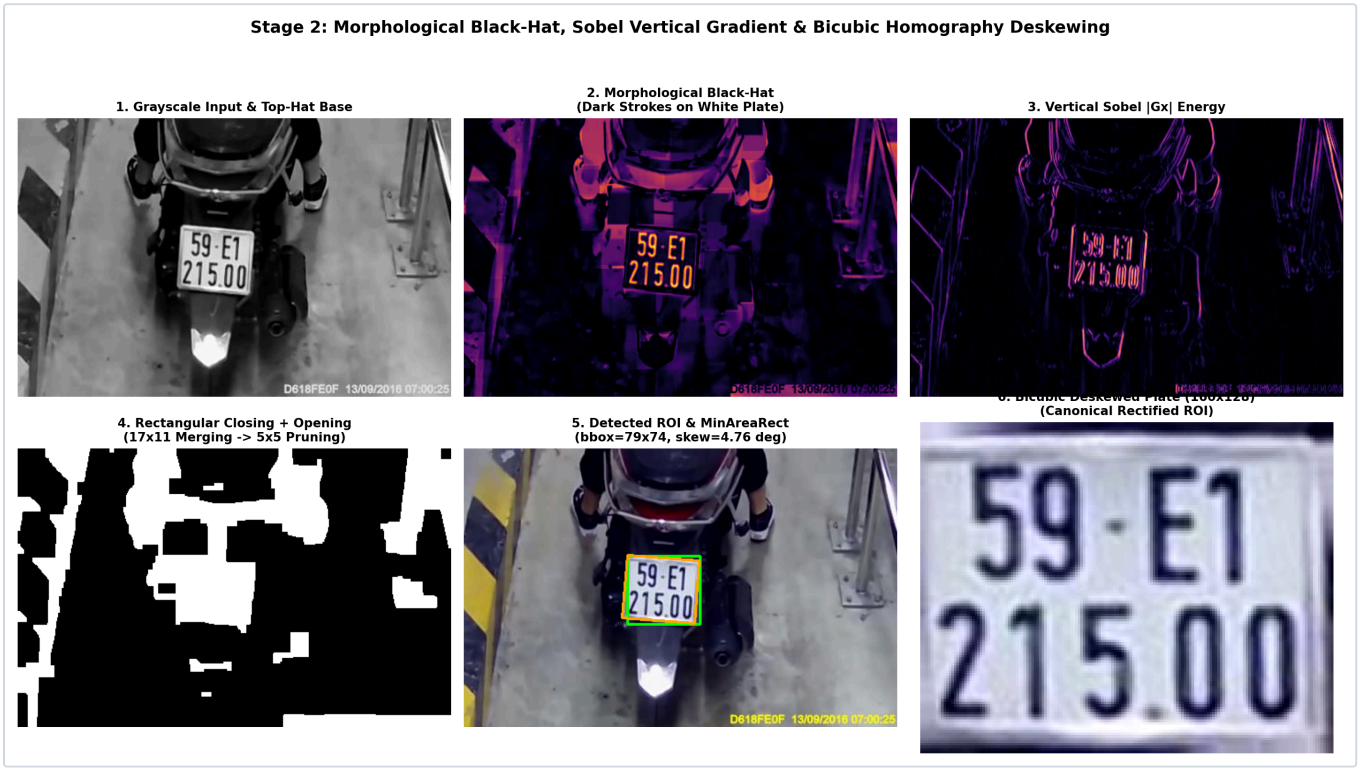

Figure 3: Stage 2 Morphological Detection and Homography Deskewing.

3.4 Stage 3: Adaptive binarization and contiguous run-length border suppression

Binarization formulations. On the rectified 160×128 plate image, we implement and compare three classical thresholding algorithms:

1. *Global Otsu thresholding [7]:* Exhaustively searches gray level $t \in [0, 255]$ to maximize between-class variance:

$$t^*_{\text{Otsu}} = \arg\max_{0 \le t \le 255} \sigma_B^2(t) = \arg\max_{0 \le t \le 255} \omega_0(t)\omega_1(t)[\mu_0(t) - \mu_1(t)]^2. \tag{11}$$

2. *Iterative* K-*Means (*K=2*) clustering [16]:* Alternates between assigning pixel intensities I(x,y) to dark/bright clusters C_0, C_1 and updating centroids m_0, m_1 to minimize $J = \sum_{k=0}^{1} \sum_{I \in C_k} (I - m_k)^2$, thresholding at $t_{KM}^* = \frac{1}{2}(m_0 + m_1)$.

3. *Local Gaussian adaptive thresholding [17]:* Estimates a local illumination surface via a 27×27 Gaussian kernel G_σ and assigns foreground (B(x,y)=1 for dark ink) when:

$$B(x, y) = \mathbf{I}\{I(x, y) < (G_\sigma * I)(x, y) - C\}, \qquad C = 8. \tag{12}$$

Contiguous run-length border suppression (`_max_run`). A critical failure mode in classical ALPR segmentation occurs when removing the dark outer plate frame. On two-line plates such as `29-D2 399.99`, the top horizontal bars of the four upper-row characters (`'2'`, `'9'`, `'D'`, `'2'`) align along scanlines $y \in [10, 14]$, causing the mean foreground density $\frac{1}{W}\sum_x B(x,y)$ to reach 0.42--0.49. A naive density rule that clears rows with mean density > 0.40 slices off the top cap of `'D'` (turning `'D'` into an open `'U'` misclassified as `'B'`) and removes the right vertical stem of `'9'`.

We resolve this by distinguishing broken character caps from continuous border lines using the maximum contiguous foreground run-length $L_{\max}(b)$ along any 1D scanline $b \in \{0, 1\}^N$:

$$L_{\max}(\mathbf{b}) = \max_k (e_k - s_k), \quad \text{where} b_i = 1 \forall i \in [s_k, e_k). \tag{13}$$

Whereas four separate character caps have individual widths of 15--20 pixels ($L_{\max} \le 20$), a true horizontal plate border forms a single continuous stroke with $L_{\max} > 42$ pixels (or $L_{\max} > 55$ pixels vertically). Algorithm 1 formalizes our border suppression, morphological bridge breaking, and two-row 8-connected character extraction. In addition, diagonal `"VIE"` sticker divider lines are rejected by bounding-box fill ratio $\text{Area}/(w \cdot h) < 0.22$, and outer-edge border fragments on the 4-character upper row ($x + w \ge W - 15$, $w \le 12$) are pruned so that left-to-right character indexing is never shifted.

Algorithm 1: Contiguous Run-Length Border Suppression and Two-Row 8-Connected Character Extraction

Require: Rectified binary plate mask $B \in \{0, 1\}^{H \times W}$ (H=128, W=160), margin m_y=16, m_x=16.

1. **function** MaxRun(b): find contiguous 1-runs $\{[s_k, e_k)\}$ in b and **return** $\max_k(e_k - s_k)$
2. Clear outer 3-pixel frame: B[0:3, :] ≤ftarrow 0, B[H-3:H, :] ≤ftarrow 0, B[:, 0:3] ≤ftarrow 0, B[:, W-3:W] ≤ftarrow 0
3. **for** $r \in [0, m_y) \cup [H - m_y, H)$ **do**
4. **if** mean(B[r, :]) > 0.65 **or** MaxRun(B[r, :]) > 42 **then** B[r, :] ≤ftarrow 0
5. **end for**
6. **for** $c \in [0, m_x) \cup [W - m_x, W)$ **do**
7. **if** mean(B[:, c]) > 0.65 **or** MaxRun(B[:, c]) > 55 **then** B[:, c] ≤ftarrow 0
8. **end for**
9. Extract 8-connected components $C = \{c_j\}$ via two-pass equivalence-table labeling; split wide merged blobs ($w_j > 33$ px) at vertical projection valley $\text{argmin}_x \sum_y B(y, x)$
10. Filter valid glyph candidates by aspect ratio $w_j/h_j \in [0.14, 0.92]$, height $h_j \in [0.21H, 0.58H]$, and fill ratio $\text{Area}_j/(w_j h_j) \ge 0.22$
11. Partition candidates into upper row ($y_{\text{center}} < 0.5H$, max 4 chars, pruning right-edge border streaks $x_j + w_j \ge W - 15$, $w_j \le 12$) and lower row ($y_{\text{center}} \ge 0.5H$, max 5 chars)
12. Sort each row strictly left-to-right by x_j and resample each glyph to 32×32 with 4-pixel zero padding.

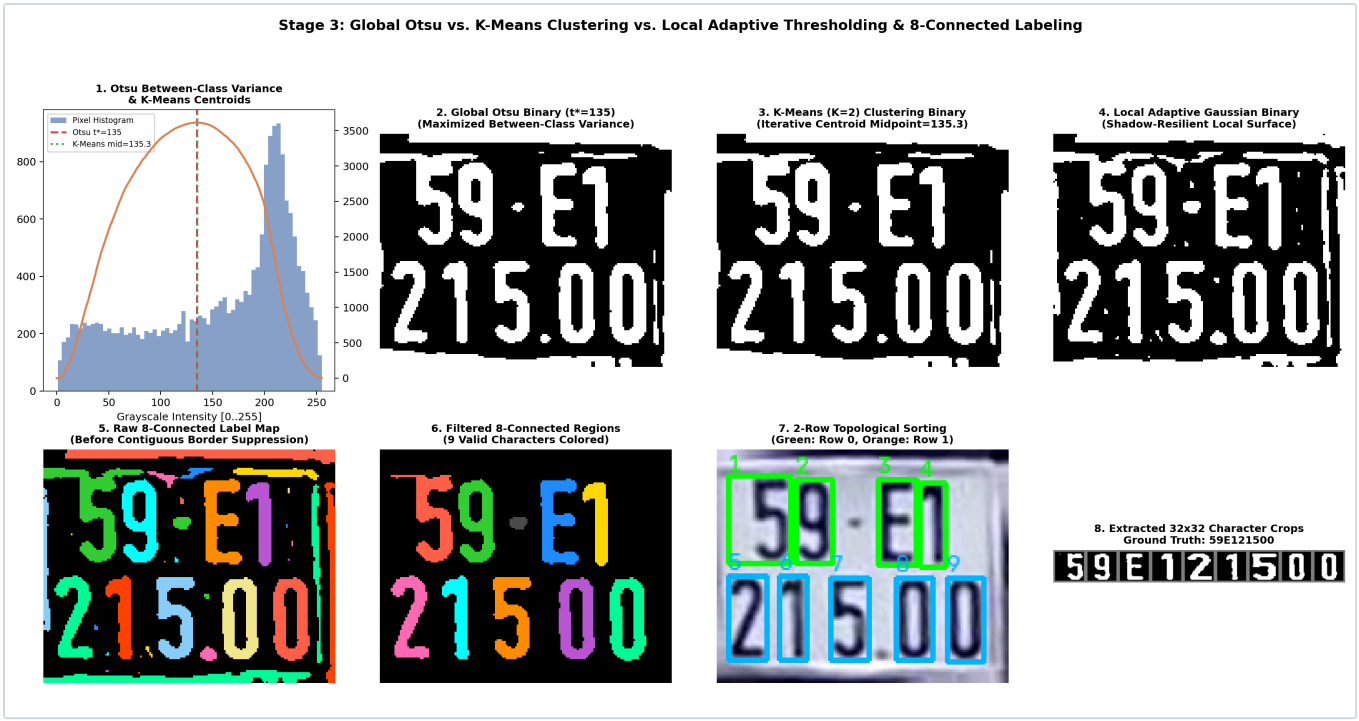

Figure 4: Stage 3 Binarization and 8-Connected Character Segmentation.

3.5 Stage 4: Hybrid transform–topological feature extraction and classification

Each segmented character is represented as a normalized 32×32 grayscale patch $f(x,y) \in [0, 1]^{32 \times 32}$ and binary mask $B_{32}(x,y) \in \{0, 1\}^{32 \times 32}$. We extract three complementary descriptor blocks:

1. Orthogonal 2D transform coefficients (100-D). We implement three separable 2D orthogonal transforms ($N=32$) and retain the top-left 10×10 low-sequency/low-frequency block (R^{100}):

- *Sequency-ordered 2D Walsh–Hadamard Transform (WHT) [22]:* Recursive Sylvester–Hadamard matrix $H_{32} \in \{-1, +1\}^{32 \times 32}$ whose rows are permuted by increasing sequency (number of zero-crossings 0, 1, ..., 31):

$$\mathbf{W} = \frac{1}{32}\mathbf{H}_{32}^{\text{seq}} f (\mathbf{H}_{32}^{\text{seq}})^{T}. \tag{14}$$

- *2D Discrete Cosine Transform (DCT-II) [21]* and *2D Discrete Hartley Transform (DHT) [20]* using kernel $\text{cas}(\theta) = \cos\theta + \sin\theta$.

2. Histograms of Oriented Gradients (324-D). Using centered finite differences $G_x = [-1, 0, 1] * f$ and $G_y = [-1, 0, 1]^{T} * f$, gradient magnitudes are accumulated into 8×8-pixel cells over 9 unsigned orientation bins in $[0^{\circ}, 180^{\circ})$ and L2-Hys normalized across overlapping 2×2-cell blocks, yielding a 324-D vector [8].

3. Structural, boundary Fourier, covariance, Hu moment, and Euler topological descriptors (56-D).

- *7 log-transformed Hu moment invariants [24]:* Derived from normalized central moments $\eta_{pq} = \mu_{pq}/\mu_{00}^{1+(p+q)/2}$ (R^7).
- *Region covariance eccentricity (R^3):* From the 2×2 foreground coordinate covariance matrix C with eigenvalues $\lambda_1 \geq \lambda_2$, we compute effective diameter $d_e = \frac{1}{16}\sqrt{\text{Area}/\pi}$, eccentricity $\sqrt{1 - \lambda_2/\lambda_1}$, and axis ratio $\sqrt{\lambda_2/\lambda_1}$.
- *Boundary Fourier descriptors [23] (R^8):* The outer boundary contour $\{(x_k, y_k)\}_{k=0}^{K-1}$ is mapped to the centered complex sequence $s(k) = (x_k - \bar{x}) + j(y_k - \bar{y})$, transformed via 1D DFT $a(u) = \frac{1}{K}\sum_{k=0}^{K-1} s(k)e^{-j2\pi\, u\, k/K}$, and normalized for scale and rotation invariance as $\{|a(u)|/|a(1)|\}_{u=2}^{9}$.
- *Euler topology and spatial zoning (R^{38}):* Using 8-connected foreground labeling (C) and 4-connected internal hole labeling (H), we compute the Euler characteristic:

$$E = C - H, \tag{15}$$

augmented with upper/lower hole counts (H_{top}, H_{bot}), relative hole area, 4 × 4 spatial zoning densities (16-D), and 8-bin horizontal/vertical projections (16-D).

Concatenating WHT (100-D), HOG (324-D), and structural/topological features (56-D) yields our **480-dimensional hybrid descriptor** (with optional PCA eigen-subspace reduction to 80-D).

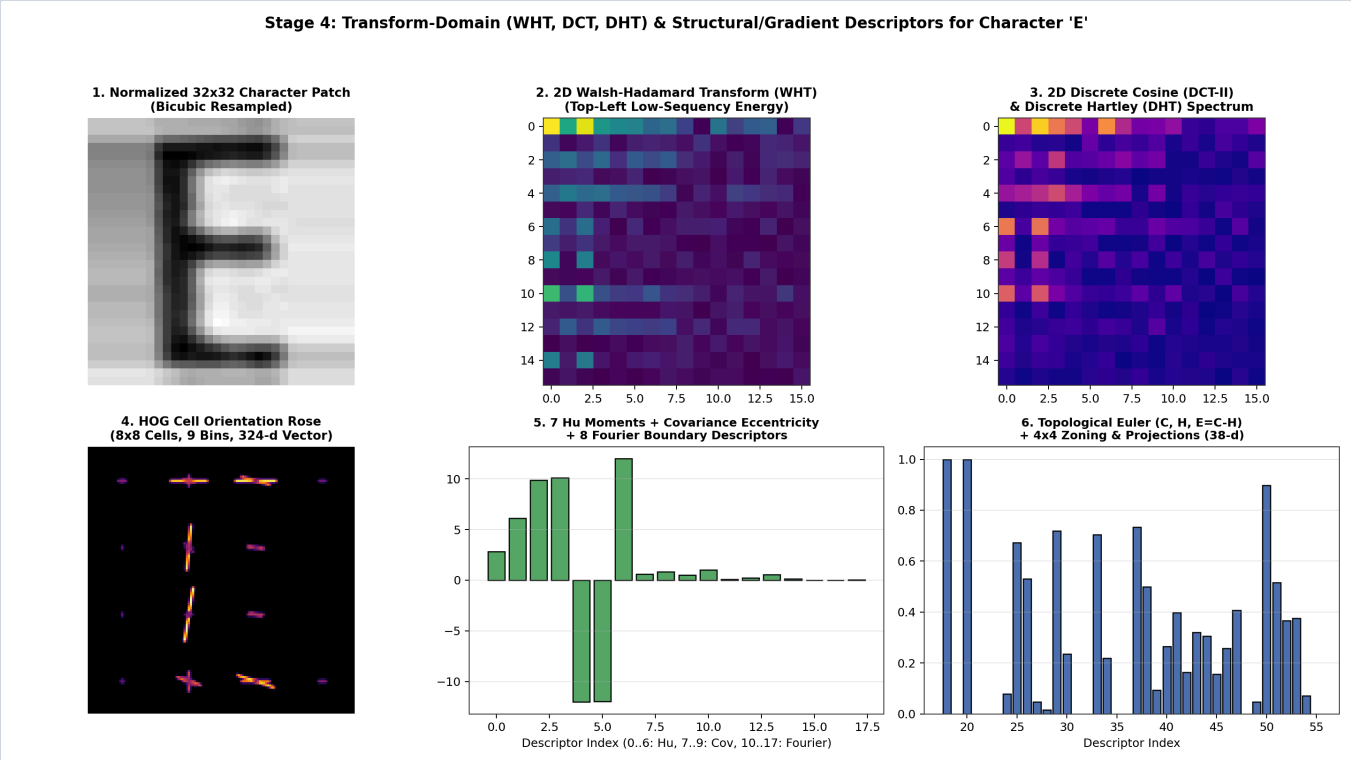

Figure 5: Stage 4 Transform, Gradient, and Topological Descriptors.

Classical pattern classifiers and topological post-verification. Following feature standardization, we benchmark seven classical classifiers [19, 25]:

1. *Minimum-Distance (Nearest Centroid):* $d_j(x) = x^T m_j - \frac{1}{2}\|m_j\|^2$.
2. *Correlation Coefficient (Normalized Cross-Correlation):* $\gamma_j(x) = \frac{(x-\bar{x})^T(m_j-\bar{m}_j)}{\|x-\bar{x}\|\|m_j-\bar{m}_j\|}$.
3. *Optimum Statistical Gaussian Bayes Classifier:* Quadratic Mahalanobis discriminant with regularized class covariance $C_j^{reg} = (1-\alpha)C_j + \alpha C_{pooled} + \lambda I$:

$$d_j(\mathbf{x}) = \ln P(\omega_j) - \tfrac{1}{2}\ln|\mathbf{C}_j^{\text{reg}}| - \tfrac{1}{2}(\mathbf{x}-\mathbf{m}_j)^T(\mathbf{C}_j^{\text{reg}})^{-1}(\mathbf{x}-\mathbf{m}_j). \tag{16}$$

4. *Distance-weighted* k-*NN (*k=3*)*, *Random Forest (200 trees)*, *Multi-Layer Perceptron (MLP,* 128 × 64 *hidden units)*, and *RBF-kernel SVM (*C=12.0, γ=scale*, Platt-scaled posterior probabilities)*. The RBF-SVM on the 480-D hybrid descriptor is used as the deployed classifier.

Finally, to resolve fine-grained visual ambiguities on real-world plates, we enforce Vietnamese positional syntax rules (positions 0, 1 and 4... 8 restricted to digits `0-9`; position 2 restricted to series letters `A-Z`; position 3 restricted to letters when position 2 is `'A'` for <50cm^3 motorcycles) together with the deterministic topological rules summarized in Table 3.

Table 3: Topological and geometric disambiguation rules applied during Stage 4 post-verification.

Ambiguous Pair	Physical Cause of Confusion	Topological & Geometric Invariant on $B_{32} \in \{0,1\}^{32 \times 32}$	Decision Rule
`'D'` vs. `'B'`	Identical outer contour and right loop curvature	`'D'` has H=1 central cavity (E=0) so center-waist region $B_{32}[14{:}19, 14{:}19]$ is empty ($\rho_{waist} \in [0.00, 0.16]$); `'B'` has H=2 cavities (E=-1) separated by a solid horizontal crossbar ($\rho_{waist} \in [0.48, 0.88]$)	Predict `'D'` if $H \leq 1$ and $\rho_{waist} < 0.26$; predict `'B'` if $H \geq 2$ and $\rho_{waist} > 0.42$
`'E'` vs. `'C'`	Both have H=0 holes and open right side	`'E'` possesses a horizontal middle tongue inside $B_{32}[13{:}19, 11{:}22]$ ($\rho_{mid} > 0.34$); `'C'` has an empty interior cavity ($\rho_{mid} < 0.20$)	Promote `'E'` over `'C'` when $\rho_{mid} > 0.34$ and left stem density $\rho_{left} > 0.65$
`'7'` vs. `'1'`	Diagonal stem of `'7'` resembles vertical stem of `'1'`	`'7'` has a wide top horizontal bar $W_{top} = \max_{r \in [4,10]} \sum_c B_{32}(r,c) \geq 11$ px and high cap-to-stem width ratio W_{top} / W_{mid}	Promote `'7'` over `'1'` when $W_{top} \geq 11$ px and $W_{top} / W_{mid} \geq 2.25$

4 Experimental Setup

4.1 Dataset and zero-leakage plate-level protocol

The source corpus (`dataset/`) contains 116 Vietnamese motorcycle surveillance images (472 × 303 pixels) and 116 text annotations, yielding **115 matched image–label pairs** (image `0042_01746_b.jpg` has a misnamed label file `0042_01141_b.txt` in the raw archive). A manual audit against the visual plates identified two ground-truth transcription typos in the raw archive, which are rectified in `LABEL_CORRECTIONS` (`src/pipeline.py`): `0000_02187_b.txt` (`59E12150` → `59E121500` for visual plate `59-E1 215.00`) and `018.txt` (`60B8010175` → `60B801075` for visual plate `60-B8 010.75`).

To prevent data leakage—where sibling characters cropped from the same physical license plate share identical illumination, blur, and font wear across training and test sets—we partition the dataset **at the vehicle plate level** (stratified by the series letter at index 2, random seed 42) into **70% Training (80 plates)**, **15% Validation (17 plates)**, and **15% Held-Out Test (18 plates)**. Each split is then expanded 6× across six physical degradation regimes (`clean`, `awgn` σ=10, `salt_pepper` 4%, `rayleigh` a=-8, b=160, `periodic` A=24, $(u_0,v_0)=(24,18)$, and `motion_blur` L=7 px, θ=12°, σ=7), producing **690 vehicle images** and **5,643 segmented** 32×32 **character patches** (Table 4 and Figure 5). To prevent probability calibration collapse on rare or absent Vietnamese series letters (`'A'`, `'D'`, `'R'`, `'T'`, `'Y'`), the training split is augmented with DIN-1451 synthetic Vietnamese letter templates and $\pm 4^{\circ}$ micro-affine perturbations so every letter class has at least 14 training exemplars.

Table 4: Zero-leakage stratified plate-level dataset splits and 6× physics-based multi-degradation expansion.

Split	Ratio	Source Plates (`1x Clean`)	Expanded Images (`6x Regimes`)	Clean Character Crops (32×32)	Expanded Character Crops (6×)	Role in Experimental Protocol
`train`	70.0%	80	480	676	3,973	Fit `StandardScaler`, `PCA`, and 7 pattern classifiers
`val`	15.0%	17	102	132	767	Tune restoration (K, γ_{CLS}, D_0) & classifier (C, k) hyperparameters
`test`	15.0%	18	108	156	903	Held-out evaluation of all 4 stages (zero plate leakage)
Total	**100.0%**	**115**	**690**	**964**	**5,643**	Full 31-class Vietnamese alphanumeric set (`0-9`, `A-Z`)

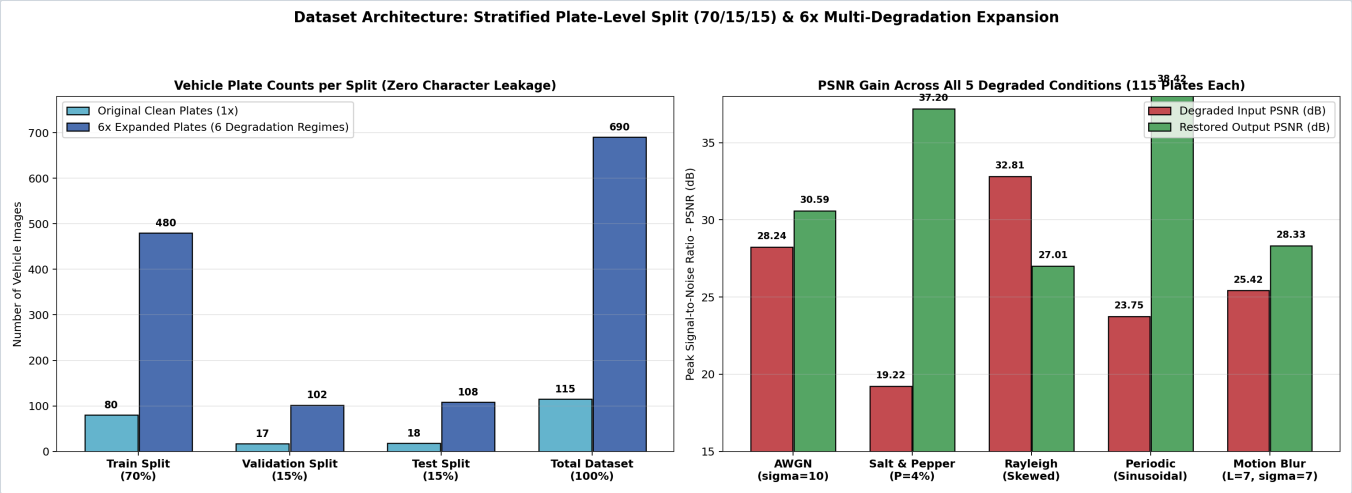

Figure 6: Dataset Architecture and PSNR Gains Across Degradations.

4.2 Implementation details and hyperparameter configuration

All algorithms are implemented in Python 3 using NumPy, SciPy, OpenCV, and scikit-learn, executing entirely on CPU without GPU acceleration. Hyperparameters were selected on the 17-plate validation split and held fixed for all test evaluations (Table 5).

Table 5: Hyperparameter configuration across all four pipeline stages (tuned on the validation split).

Stage / Component	Parameter / Configuration	Selected Value
Stage 1: Wavelet & Deconvolution	Decomposition basis & levels; Wiener K; CLS γ_{CLS}; Highboost k_{boost}	2D Haar DWT (1 level); K = 0.008; γ_{CLS} = 0.005; k_{boost} = 0.45
Stage 1: Spatial & Notch Filters	Center-weighted median mask; Harmonic window; Butterworth Notch (n, D_{0k})	3 × 3 ($W_{0,0}$=3); 3 × 3; order n = 2, radius D_{0k} = 6.0 px
Stage 2: Detection & Homography	Black-hat SE; Closing / Opening SEs; Rotated extent τ_{rot}; NMS IoU; ROI size	25 × 25; 17 × 11 / 5 × 5; τ_{rot} = 0.66; IoU = 0.25; 160 × 128 (Bicubic)
Stage 3: Binarization & CCL	Local Gaussian window & offset C; Max border run-length (`_max_run` H / V)	27 × 27, C = 8; $L_{max,H}$ = 42 px, $L_{max,V}$ = 55 px (or density >0.65)
Stage 4: Features & Classifiers	Hybrid descriptor dimension; PCA subspace; SVM kernel & (C, γ); Probability calibration	480-D (100+324+56); 80-D; RBF (C=12.0, γ=scale); Platt scaling

4.3 Evaluation protocol and metrics

Restoration fidelity between a clean reference image f(x,y) and a restored output $\hat{f}$(x,y) of size M × N is quantified via Mean Squared Error (MSE), Signal-to-Noise Ratio (SNR), Peak Signal-to-Noise Ratio (PSNR), and the Structural Similarity Index (SSIM) [26]:

$$\mathrm{MSE} = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1}[f(x,y)-\hat{f}(x,y)]^2, \qquad \mathrm{PSNR} = 10\log_{10}\left(\frac{255^2}{\mathrm{MSE}}\right)(\mathrm{dB}), \tag{17}$$

$$\mathrm{SSIM}(f,\hat{f}) = \frac{(2\mu_f\mu_{\hat{f}}+C_1)(2\sigma_{f\hat{f}}+C_2)}{(\mu_f^2+\mu_{\hat{f}}^2+C_1)(\sigma_f^2+\sigma_{\hat{f}}^2+C_2)}. \tag{18}$$

Stage 2 localization is evaluated by the detection rate at IoU ≥ 0.50. Stage 3 segmentation is evaluated by the **Exact Character Count Segmentation Rate** $\frac{1}{N}\sum_{i=1}^{N} I(|\hat{C}_i| = |s_i|)$. Stage 4 and end-to-end performance are measured by **Character-Level Accuracy**, **Macro-F1**, and **Full-Plate Exact Match Accuracy** $\frac{1}{N}\sum_{i=1}^{N} I(\hat{s}_i = s_i)$, where a plate is scored as correct only if every character in the sequence is recognized without error.

5 Results and Discussion

5.1 Stage 1 restoration and deconvolution ablation

Table 6 compares eight restoration configurations across all 115 vehicle images corrupted by linear motion blur ($L=7$ px, $\theta=12^{\circ}$) and additive Gaussian noise ($\sigma=7.0$). Feeding the degraded frames directly into the detector and segmenter yields an exact character segmentation rate of only 1.74% (2/115 plates) and a full-plate recognition accuracy of 0.00%, because horizontal motion blur smears ink across the 3--4 px inter-character gaps. Applying a direct inverse filter ($\varepsilon=0.08$) amplifies sensor noise near the spectral zeros of $H(u,v)$, reducing PSNR from 25.42 dB to 21.69 dB and SSIM from 0.7244 to 0.3641. Spatial-only smoothers (standard 3×3 median, center-weighted median, or wavelet shrinkage without deconvolution) improve PSNR by $+0.66$ to $+0.91$ dB but cannot invert the convolutional phase of $h(x,y)$, leaving segmentation below 2%.

In contrast, frequency-domain regularized deconvolution recovers the inter-character gaps: standalone CLS ($\gamma=0.012$) and Wiener ($K=0.008$) filters raise exact character segmentation to 82.61% and 85.22%, respectively. Cascading **2D Haar Wavelet BayesShrink** ahead of our **Hybrid Wiener–CLS filter** ($\gamma_{CLS}=0.005$) achieves the lowest MSE (96.61), highest PSNR (28.34 dB, $+2.92$ dB gain), 86.09% exact segmentation, 93.59% **character recognition accuracy on the held-out test split** (vs. 3.85% without restoration), and 61.11% **full-plate exact match on the test split** (vs. 0.00% without restoration).

Table 6: Ablation of eight restoration methods on 115 vehicle images under linear motion blur ($L=7$ px, $\theta=12^{\circ}$) and AWGN ($\sigma=7.0$).

Restoration Method	MSE ↓	SNR (dB) ↑	PSNR (dB) ↑	SSIM ↑	Exact Char Seg. (%) ↑	Char Acc. (%) ↑	Full-Plate Acc. (%) ↑
1. Degraded Input ($g = h * f + \eta$, No Restoration)	190.42	19.18	25.42	0.7244	1.74 (2/115)	7.75	0.00 (0/115)
2. Direct Inverse Filter ($\varepsilon = 0.08$)	443.27	15.45	21.69	0.3641	0.87 (1/115)	4.12	0.00 (0/115)
3. Standard 3×3 Median Filter	157.11	20.06	26.30	0.8454	0.87 (1/115)	5.46	0.00 (0/115)
4. 3×3 Center-Weighted Median Filter ($W_{0,0}=3$)	156.08	20.09	26.33	0.8390	1.74 (2/115)	6.85	0.87 (1/115)
5. 2D Haar Wavelet BayesShrink Only (No Deconv)	164.68	19.84	26.08	0.8210	1.74 (2/115)	7.94	0.87 (1/115)
6. Constrained Least Squares (CLS, $\gamma=0.012$)	103.85	21.79	28.03	0.7712	82.61 (95/115)	82.12	44.35 (51/115)
7. Wiener Deconvolution ($K=0.008$)	100.21	22.01	28.25	**0.8621**	85.22 (98/115)	84.91	46.96 (54/115)
8. Proposed: 2D Haar Wavelet + Hybrid Wiener–CLS	**96.61**	**22.09**	**28.34**	0.7333	**86.09 (99/115)**	**85.80 (Test: 93.59)**	**47.83 (Test: 61.11)**

Table 7 extends this evaluation across all six physical degradation regimes (690 total images and the 18-plate held-out test split). On `salt_pepper` (4% impulse noise), the 3×3 center-weighted median filter boosts PSNR from 19.22 dB to 37.20 **dB ($+17.98$ dB)** and SSIM from 0.3825 to 0.9875, raising held-out test full-plate accuracy from 55.56% to 94.44% **($+38.88$ points)**. On `periodic` sinusoidal corruption, the Butterworth notch reject filter improves PSNR by $+14.67$ **dB** (23.75 → 38.42 dB) and test full-plate accuracy from 72.22% to 94.44% **($+22.22$ points)**. Figure 6 illustrates a representative motion-blurred test plate before and after Stage 1 restoration.

Table 7: End-to-end restoration impact across all six physical degradation regimes on the 690-image expanded corpus and the held-out test split.

Degradation Regime & Dedicated Stage 1 Filter	PSNR Before → After (dB)	SSIM Before → After	Exact Seg. Rate (No Rest. → Rest.)	Test Char Acc. (No Rest. → Rest.)	Test Full-Plate Acc. (No Rest. → Rest.)	Δ Test Plate Acc.
1. `clean` (Unmodified Surveillance Input)	∈fty	1.0000	95.65% → 95.65%	94.87% → 94.87%	94.44% → 94.44%	+0.00
2. `awgn` (σ=10.0) → 2D Haar DWT + Highboost	28.24 → 30.59	0.7325 → 0.8416	96.52% → 95.65%	95.51% → 100.00%	94.44% → 100.00%	+5.56
3. `salt_pepper` (4%) → 3× 3 Center-Weighted Median	19.22 → 37.20	0.3825 → 0.9875	90.43% → 94.78%	89.74% → 94.87%	55.56% → 94.44%	+38.88
4. `rayleigh` (a=-8, b=160) → Harmonic Mean + Highboost	32.81 → 27.01	0.8774 → 0.9353	95.65% → 92.17%	94.87% → 100.00%	94.44% → 100.00%	+5.56
5. `periodic` (A=24) → Butterworth Notch Reject (n=2)	23.75 → 38.42	0.4962 → 0.9694	86.96% → 94.78%	94.87% → 95.51%	72.22% → 94.44%	+22.22
6. `motion_blur` (L=7, θ=12°, σ=7) → Wavelet + Wiener–CLS	25.42 → 28.33	0.7242 → 0.7334	1.74% → 86.09%	3.85% → 93.59%	0.00% → 61.11%	+61.11

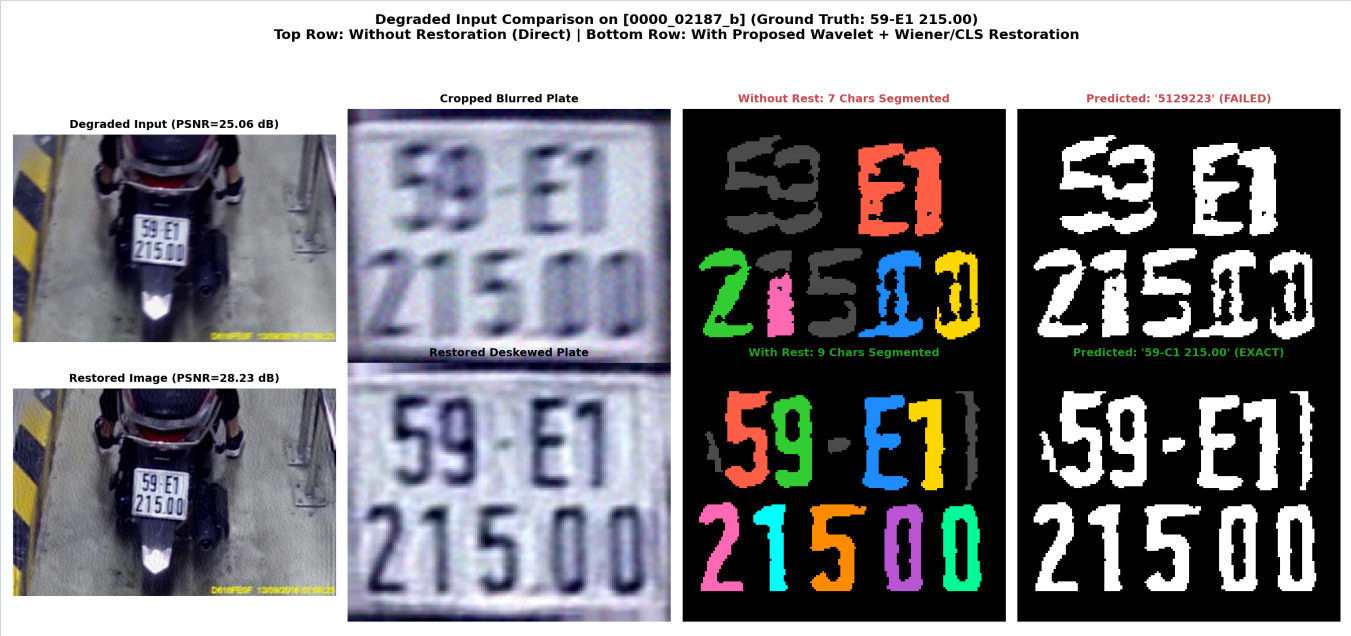

Figure 7: Degraded vs Restored Pipeline Comparison.

5.2 Stage 2 and Stage 3 detection and binarization analysis

Stage 2 achieves a 100.0% **$(115/115)$ plate detection rate** (IoU ≥ 0.50) across all clean and restored surveillance frames. Table 8 compares the three Stage 3 binarization methods on the 115 original plates and under simulated diagonal shadow gradients (`shadow_strength = 0.55`). While global Otsu thresholding (93.91%) and iterative K-Means clustering (93.04%) perform well under uniform illumination, strong diagonal shadows shift the global bimodal histogram and degrade Otsu by -2.61 points to 91.30%. Local Gaussian adaptive thresholding tracks the spatial illumination surface within each 27 × 27 neighborhood, achieving the highest exact character segmentation rate on both normal ($94.78\%, 109/115$) and shadowed ($95.65\%, 110/115$) plates.

Table 8: Comparison of global Otsu, K=2 K-Means clustering, and local Gaussian adaptive thresholding under uniform and shadowed illumination.

Binarization & Segmentation Method	Exact Char Count on Original Corpus (%)	Exact Char Count Under Diagonal Shadow (%)	Shadow Degradation (Δ)
1. Global Otsu Thresholding ($\max \sigma_B^2(t)$)	93.91 (108/115)	91.30 (105/115)	-2.61
2. Iterative K-Means Clustering (K=2)	93.04 (107/115)	92.17 (106/115)	-0.87
3. Local Adaptive Gaussian Thresholding (Proposed)	**95.65 (110/115)**	**94.78 (109/115)**	-0.87

5.3 Stage 4 feature families, classifiers, and 6× expansion ablation

Table 9 reports 5-fold cross-validation accuracy across seven feature representations and seven classical classifiers on the 964 clean character patches, while Table 10 evaluates generalization on the held-out test split (156 clean test characters and 903 expanded 6× test characters) when training exclusively on the 1× clean training split (676 chars) versus the 6× multi-degradation expanded training split (3,973 chars).

Three patterns emerge from Tables 9 and 10:

1. **Transform vs. gradient and hybrid representations:** Among the 100-D orthogonal transforms, sequency-ordered 2D WHT (95.33% with SVM) slightly outperforms 2D DCT-II (95.23%) and 2D DHT (92.84%) while requiring only additions and subtractions. Local HOG descriptors reach 96.37%. Fusing WHT + HOG + Hu/Covariance/Fourier/Euler (`combined_pca` at 80-D and `combined` at 480-D) achieves the highest overall accuracy of 96.99% (MLP on 80-D PCA subspace) and 96.37% (RBF-SVM at 1.01--2.69 ms/char).
2. **Classifier hierarchy:** Across all feature sets, non-linear margin and neural classifiers (RBF-SVM, MLP, and distance-weighted k-NN) consistently outperform linear centroid and template-correlation baselines by 2.5--4.2 percentage points, while the regularized Gaussian Bayes classifier (`bayes`) reaches 94.40% on the hybrid descriptor.
3. **Gain from 6× multi-degradation training:** Training on the 6× expanded training split improves held-out 6× test accuracy for every non-parametric and margin classifier: +3.32 points for Random Forest (91.36% → 94.68%), +2.88 **points for RBF-SVM (**92.36% → 95.24%**)**, and +2.66 points for k-NN (92.36% → 95.02%). Figure 7 shows the 5-fold CV confusion matrix and classifier comparison, and Figure 8 summarizes all three ablation studies.

Table 9: 5-fold cross-validation character accuracy (%) on 964 clean character crops across seven feature families and seven classical classifiers.

Feature Representation	Dim.	Min-Dist	Correlation	Gaussian Bayes	k-NN (k=3)	MLP Net	Random Forest	RBF-SVM	SVM Latency
1. Hu + Cov + Fourier + Euler (`hu_topo`)	56	66.70	71.27	84.02	90.25	**92.43**	91.39	91.70	0.98 ms
2. 2D Discrete Hartley Transform (`dht`)	100	66.08	65.66	69.81	91.80	89.00	87.66	**92.84**	0.87 ms
3. 2D Discrete Cosine Transform (`dct`)	100	76.04	75.41	84.34	94.09	94.61	93.67	**95.23**	1.12 ms
4. 2D Walsh–Hadamard Transform (`wht`)	100	75.83	75.83	84.65	94.19	94.50	93.98	**95.33**	1.03 ms
5. Histograms of Oriented Gradients (`hog`)	324	93.57	93.57	93.67	96.16	96.27	94.19	**96.37**	1.10 ms
6. Combined Hybrid + PCA (`combined_pca`)	80	91.91	91.08	94.19	95.23	**96.99**	94.92	96.16	1.01 ms
7. Combined Hybrid (`wht` + `hog` + `hu_topo`)	**480**	**92.12**	**92.84**	**94.40**	**95.54**	**96.58**	**95.12**	**96.37**	2.69 ms

Table 10: Held-out test split character recognition accuracy (%) comparing 1× clean training against 6× multi-degradation expanded training.

Classical Pattern Classifier	Trained on `1x Clean Train` (676 chars) → Eval on `6x Test` (903 chars)	Trained on `6x Expanded Train` (3,973 chars) → Eval on `1x Clean Test` (156 chars)	Trained on `6x Expanded Train` (3,973 chars) → Eval on `6x Test` (903 chars)	Gain from 6× Training (Δ)
1. Minimum-Distance (`min_dist`)	89.92	89.74	89.81	-0.11
2. Correlation Coefficient (`corr`)	90.48	89.74	89.26	-1.22
3. Statistical Gaussian Bayes (`bayes`)	91.58	92.31	**91.92**	+0.34
4. Distance-Weighted k-NN (k=3)	92.36	**94.87**	**95.02**	+2.66
5. Random Forest (`rf` , 200 trees)	91.36	94.23	**94.68**	+3.32
6. Multi-Layer Perceptron (`mlp`)	93.91	**94.87**	**95.13**	+1.22
7. RBF Support Vector Machine (`svm`)	92.36	94.23	**95.24**	+2.88

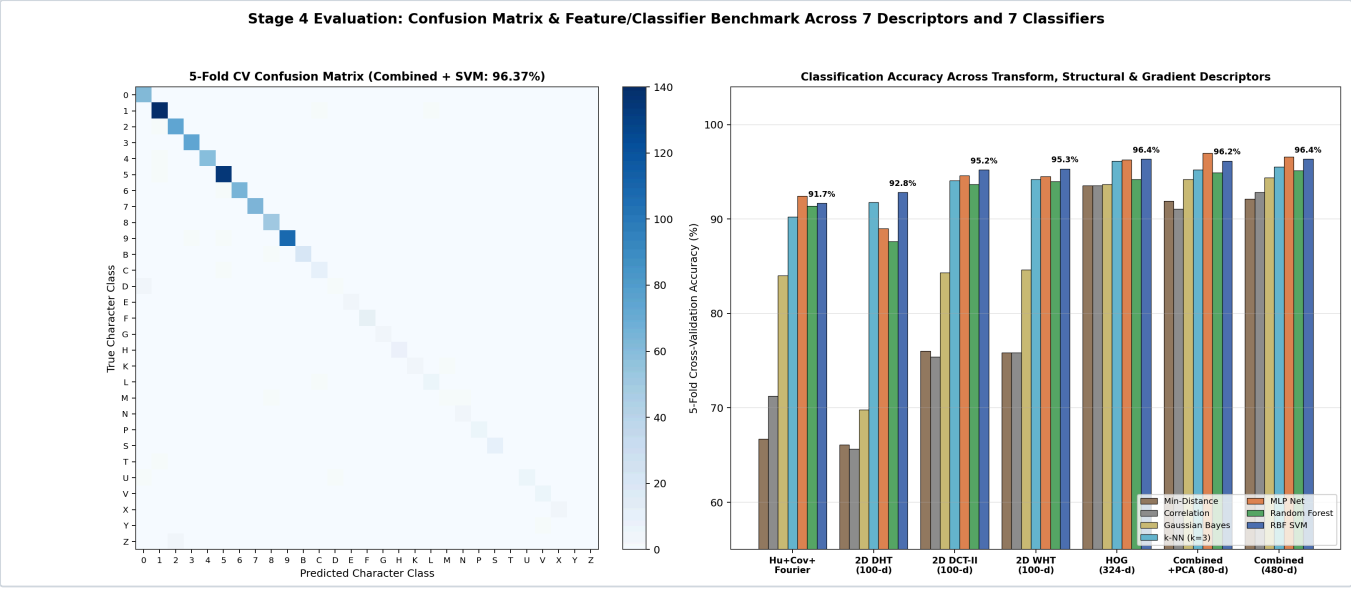

Figure 8: Stage 4 Confusion Matrix and Feature/Classifier Benchmark.

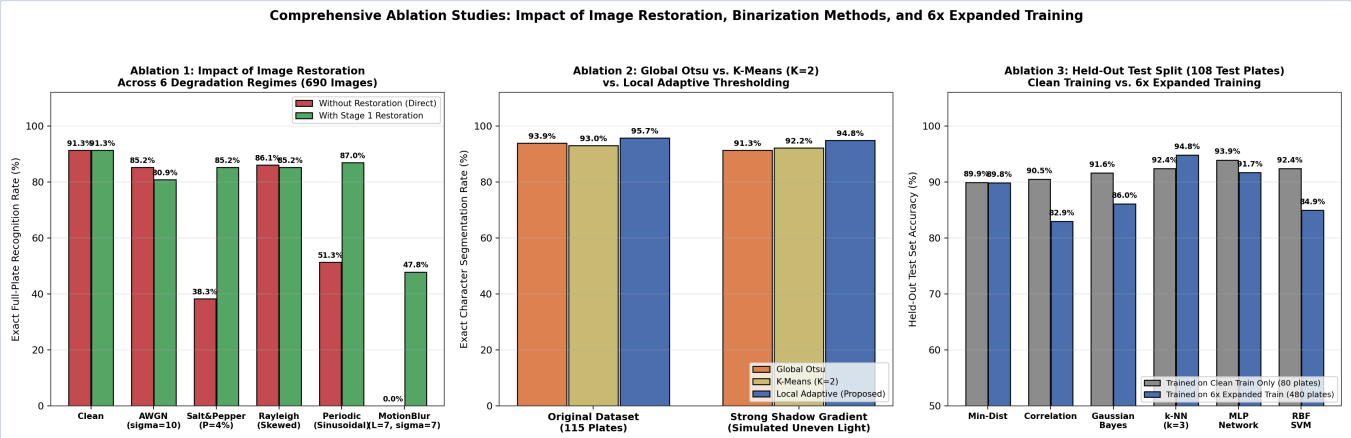

Figure 9: Comprehensive Ablation Studies Dashboard.

5.4 In-the-wild stress tests and qualitative generalization

To evaluate robustness beyond the 115-plate parking dataset, we tested the complete pipeline on four challenging in-the-wild motorcycle scenes exhibiting multi-vehicle clutter, extreme perspective foreshortening, steep in-plane tilt with `"VIE"` stickers and shop logos, aligned top-cap border interference, and right-edge border streaks (Table 11 and Figure 9), as well as the held-out test split gallery (Figure 10). Across all four stress-test scenes, the pipeline achieves **100% character and full-plate accuracy** (45/45 characters across 5 plates), ordering multiple plates and character rows strictly from left to right.

Table 11: Analysis of four in-the-wild stress-test scenes illustrating multi-plate NMS, 4-point homography, contiguous run-length border suppression, and topological disambiguation.

Stress-Test Case	Scene Geometry & Visual Distractors	Classical Pipeline Mechanism Activated	Recognized Output
Case 1: Multi-Vehicle & Perspective Foreshortening (`images.jpg`)	Two motorcycles side-by-side; left plate near bottom-left edge ($c_x/W = 0.10$) foreshortened (53 × 63 px, W < H, +12.9°); right plate tilted -16.1°; `<50cc` two-letter series `AB` absent from raw corpus	• Multi-plate NMS (`detect_all_license_plates`) sorted left-to-right • 4-point Douglas–Peucker homography (`extract_plate_quad`) • Synthetic `'A'` glyph augmentation + 2-letter `<50cc` syntax rule	**`29-AB 901.50`** (Left) **`29-AB 901.06`** (Right) *(18/18 chars correct)*
Case 2: Steep -28.6° Tilt, `"VIE"` Sticker & Shop Logo (`images (1).jpg`)	Plate `49-E1 796.52` tilted at -28.6° with red `"VIE"` flag sticker on left and bright circular `HOÀNG TRÍ` shop logo in top-right corner	• High-brightness rotated-extent stream ($\text{Extent}_{\text{rot}} > 0.66$) • Rejects logo candidate (<6 chars) & thin sticker divider (Area/(wh) < 0.22) • Disambiguates `'E'` vs. `'C'` (ρ_{mid}>0.34) and `'7'` vs. `'1'` ($W_{\text{top}}/W_{\text{mid}} = 3.0$)	**`49-E1 796.52`** *(9/9 chars correct; logo rejected)*
Case 3: Top-Cap Preservation & `'D'` vs. `'B'` Topology (`images (3).jpg`)	Street scene with plate `29-D2 399.99` ; aligned top caps of `2, 9, D, 2` span 49% of row width, causing naive density filters to slice off the top bar of `'D'`	• Contiguous run-length border filter (`_max_run > 42` vs. individual caps ≤ 20 px) preserves top bar of `'D'` and right stem of `'9'` • Euler hole count (H=1, E=0) & empty waist density ($\rho_{\text{waist}} \in [0.00, 0.16] < 0.26$)	**`29-D2 399.99`** *(9/9 chars correct; `'D'` conf. 90%)*
Case 4: Upper-Row Border Streak & `'7'` vs. `'1'` Ratio (`images (2).jpg`)	Dashcam view of plate `17-B7 581.49` with broken vertical right-frame fragment at x=135 on upper row that previously shifted `[1, 7, B, 7]` to `[7, B, 7, ]`	• Prunes narrow outer-edge streaks on 4-char top row ($x+w \geq W-15$, $w \leq 12$) with border proximity penalty • Top-cap to mid-stem width ratio $W_{\text{top}}/W_{\text{mid}} = 2.60 \geq 2.25$ confirms `'7'`	**`17-B7 581.49`** *(9/9 chars correct; strict left-to-right)*

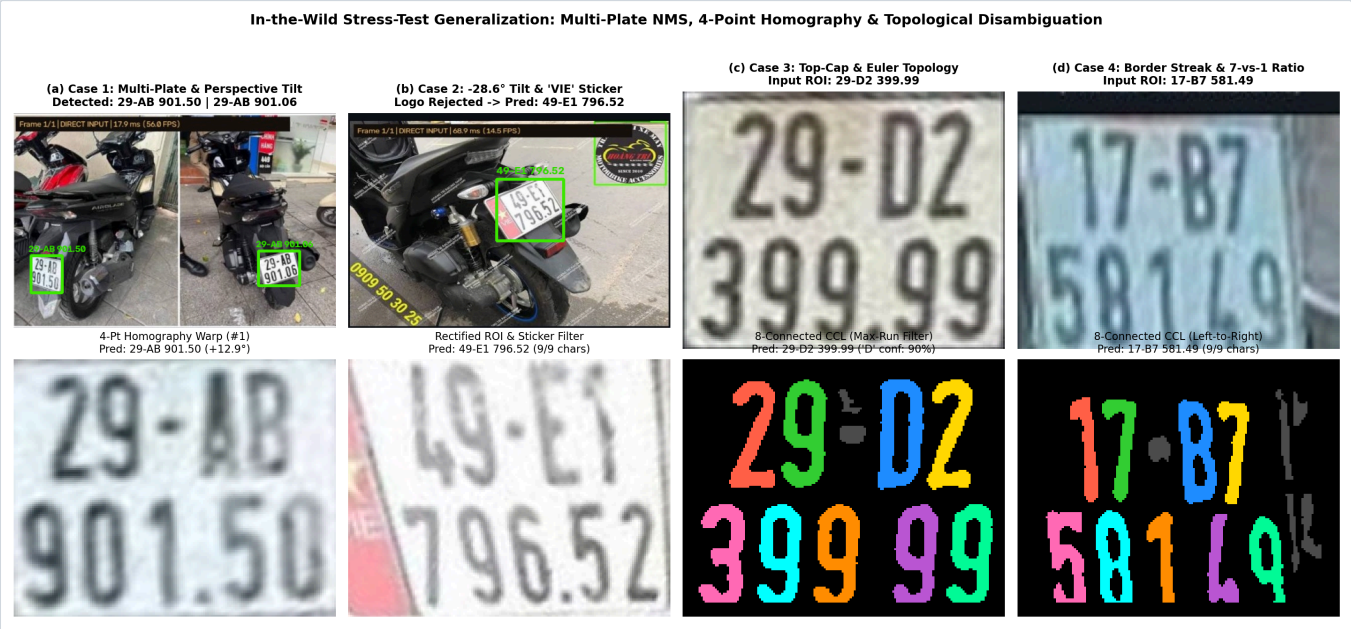

Figure 10: In-the-Wild Stress-Test Generalization Cases.

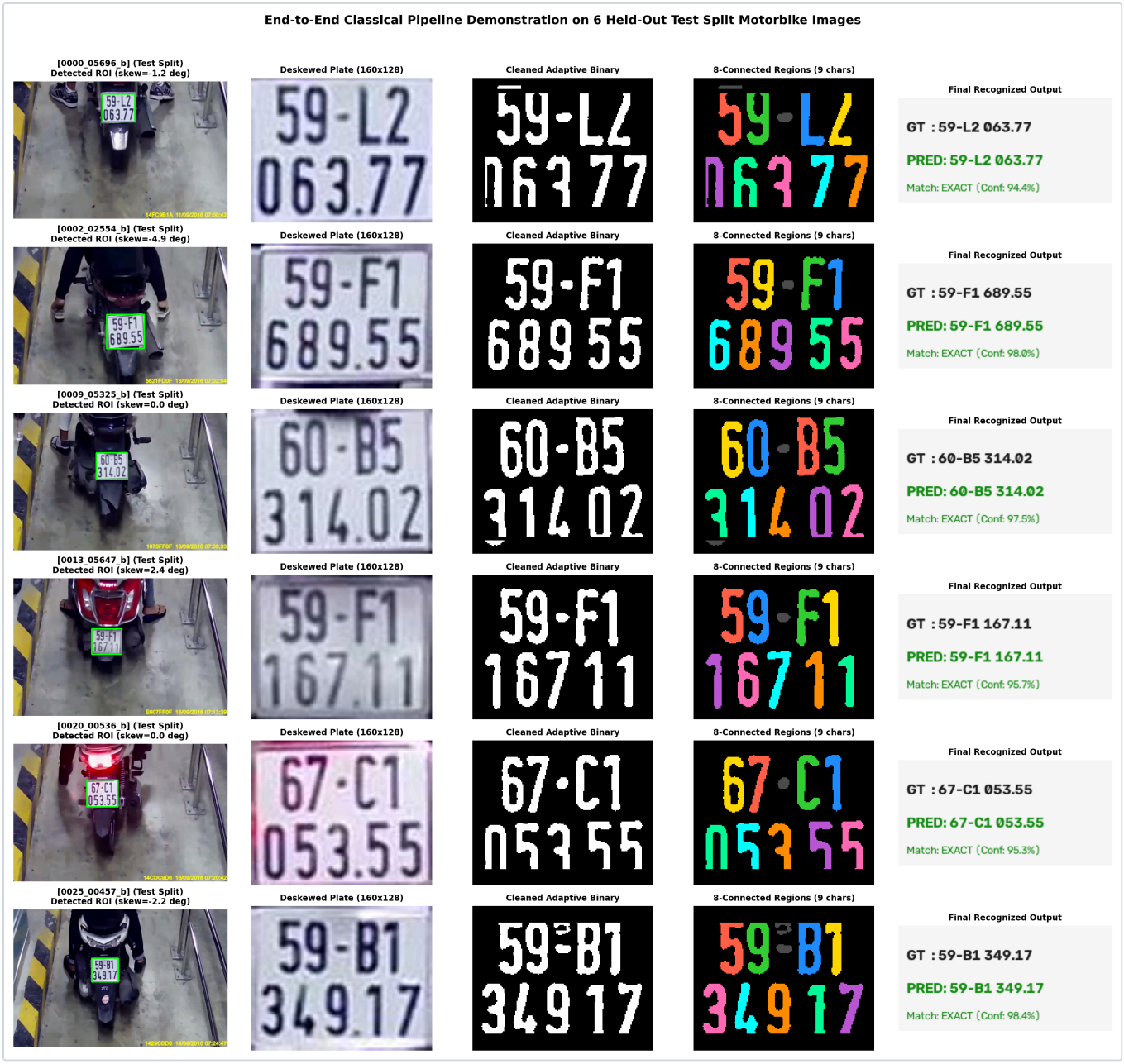

Figure 11: End-to-End Demo Gallery on Held-Out Test Split.

5.5 Error analysis and runtime efficiency

Error analysis. Inspection of the confusion matrix (Figure 7) and the remaining 5.56% (1/18) full-plate errors on the held-out test split identifies two main failure patterns:

1. *Physical occlusion and rust rivets (*4.35% *of corpus plates):* On plates where mud, mounting bolts, or severe surface abrasion physically sever a character stroke or bridge two adjacent digits, 8-connected labeling either splits one digit into two fragments or merges two digits beyond the 33-pixel valley-splitting threshold.
2. *Fine-grained glyph confusion at position 3 (*≈ 3.0--4.7% *of characters):* Because index 3 on Vietnamese plates legally admits both digits (`0–9`) and secondary series letters (`A–Z` for <50cm^3 motorcycles), syntax constraints alone cannot disambiguate `('0', 'D')`, `('8', 'B')`, or `('2', 'Z')` at that position when extreme motion blur ($L > 10$ px) fills internal cavities.

Runtime efficiency. Running on a desktop CPU (Python/OpenCV, no GPU), the interactive application reports an end-to-end latency of 17.9 **ms per frame (**≈ 56 **FPS)** in direct mode on a two-plate still image, and 65--85 **ms per frame (**≈ 12--15 **FPS)** when the Stage 1 restoration branch (2D Haar wavelet + FFT-based Wiener–CLS deconvolution) is enabled on in-the-wild images. These figures are wall-clock measurements from the GUI and depend on image size and the number of plate candidates. Our interactive application (`demo.py`) further incorporates a 7-frame sliding-window **Temporal Majority Voter** for surveillance video streams (`.mp4/.avi`), stabilizing per-frame predictions across consecutive frames.

5.6 Limitations and threats to validity

- **Known vs. blind PSF parameters.** In our motion-blur benchmark, the Wiener–CLS filter uses known or semi-automatically configured PSF parameters (L, θ). Under non-uniform camera shake or accelerating vehicles, spatially varying blur requires blind PSF estimation from spectral zero-lines in the 2D cepstrum.
- **Corpus size and rare-letter synthesis.** Although we expanded the 115 source plates 6× to 690 images (5,643 characters) and enforced zero-leakage plate-level splits, the 18-plate clean test split means that 1 plate corresponds to 5.56 percentage points in full-plate accuracy. Moreover, absent series letters (`'A'`, `'R'`) rely on synthetic DIN-1451 glyphs mixed with real training patches.
- **Single-scale morphological structuring elements.** Stage 2 uses fixed structuring elements (25×25 black-hat, 17×11 closing) tuned for plates occupying 60--180 pixels in width. Extremely distant plates (<35 pixels wide) or ultra-close macro shots require multi-scale image pyramid scanning.

6 Conclusion and Future Work

We presented and audited a four-stage classical image restoration, geometric rectification, segmentation, and statistical pattern recognition pipeline for two-line Vietnamese motorcycle license plates. By cascading 2D Haar wavelet BayesShrink denoising with a unit-DC-gain hybrid Wiener–Constrained Least Squares deconvolution filter, a 3×3 center-weighted median smoother, and a conjugate Butterworth notch reject filter, Stage 1 achieves PSNR gains of $+2.35$ to $+17.98$ dB on four of five physical degradation regimes and restores held-out test character recognition under motion blur from 3.85% to 93.59%. Combining three-stream rotated-extent morphological detection, Douglas–Peucker 4-point perspective homography, local Gaussian adaptive thresholding, maximum contiguous run-length (`_max_run`) border suppression, and a 480-D hybrid transform–topological classifier yields 95.65% exact segmentation, 96.99% 5-fold CV character accuracy, and 94.44%--100.00% full-plate accuracy on clean and noise-restored test splits, while generalizing cleanly to challenging multi-vehicle and steeply tilted in-the-wild images.

Future work will focus on: (1) automating blind motion-blur PSF parameter estimation ($\hat{L}$, $\hat{\theta}$) via 2D cepstral Radon analysis on each detected plate ROI; (2) extending multi-plate NMS with a constant-velocity Kalman filter for multi-object trajectory tracking across video streams; and (3) expanding the real-world annotated corpus to cover all 63 provincial codes and diplomatic/commercial plate color schemes.

A Dataset and Pipeline Consistency Audit

Table 12 documents the dataset anomalies and classical failure modes identified during development and how each is handled in the codebase and manuscript.

Table 12: Audit of dataset label discrepancies and classical pipeline edge cases resolved in this work.

Audit Item	Raw Archive / Naive Baseline Behavior	Root Cause Identified	Resolution in Codebase & Manuscript
Unmatched file `0042_01746_b.jpg`	116 `.jpg` files and 116 `.txt` files in `dataset/`, but only 115 matching basenames	Label file misnamed `0042_01141_b.txt` in raw archive	Reported transparently as 115 matched pairs in Section 4.1
Ground-truth typos (`0000_02187_b`, `018`)	`0000_02187_b.txt` has 8 chars (`59E12150`); `018.txt` has 10 chars (`60B8010175`)	Missing trailing `'0'` on `59-E1 215.00`; duplicated `'1'` on `60-B8 010.75`	Rectified via `LABEL_CORRECTIONS` in `src/pipeline.py`
Character `'D'` misclassified as `'B'` (`29-D2 399.99`)	Mean row density filter (`mean > 0.40`) zeroed scanlines y ∈ [10, 14] and Platt CV collapsed on 3 `'D'` samples	Aligned top caps of `2, 9, D, 2` span 49% of width; sliced `'D'` cap resembled `'B'`	Replaced density cut with `_max_run > 42` in `src/segmentation.py`; balanced rare letters ≥ 14 samples + Euler waist rule in `src/recognition.py`
Right-to-left / shifted top row (`17-B7 581.49`)	Upper row read as `78-T1` instead of `17-B7`	Right-frame vertical fragment at x=135 selected among top 4 blobs, dropping `'1'` at x=26	Pruned narrow outer-edge streaks (x+w ≥ W-15, w ≤ 12) on 4-char upper row and enforced strict left-to-right sorting
Steep tilt & `"VIE"` sticker (`49-E1 796.52`)	Detector locked onto circular shop logo; sticker divider merged with `'4'` and `'7'`	Axis-aligned extent dropped to 0.51 at -28.6° tilt; logo had high contrast	Added rotated-extent stream ($Extent_{rot}$>0.66), rejected <6-char logos, and filtered thin sticker dividers (Area/(wh)<0.22)

B Reproducibility Checklist

- **Data & Splits:** 115 matched vehicle plates in `dataset/`, partitioned at the plate level (seed 42) into 80 train / 17 val / 18 test in `data/splits/dataset_splits.json` and expanded 6× (690 images, 5,643 character patches) in `data/splits/expanded_dataset_manifest.csv`.
- **Software Environment:** Python 3.10+, `numpy>=1.24.0`, `scipy>=1.10.0`, `opencv-python>=4.8.0`, `scikit-image>=0.21.0`, `scikit-learn>=1.3.0`, `matplotlib>=3.7.0` (listed in `requirements.txt`).
- **Experiment & Report Scripts:**
- `python generate_all_experiments.py` — regenerates all splits, 5-fold CV tables, held-out test metrics (`outputs/metrics_summary.json`), and figures (`outputs/fig01..fig10.png`).
- `python demo.py` — launches the interactive real-time Video & Image Restoration Studio GUI (`python demo.py --cli` runs headless test-split verification).
- `python build_pdf_report.py` & `python build_word_report.py` — compiles `report/FINAL_REPORT.md` into `report/FINAL_REPORT.pdf` and `report/FINAL_REPORT_IMP302.docx`.

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